

TYPE I BLOW-UP LIMITS AND RESIDUAL CLOSURE FOR THE THREE-DIMENSIONAL NAVIER-STOKES EQUATIONS

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ABSTRACT. We reduce an admissible finite-energy Type I singularity for the three-dimensional Navier–Stokes equations to a normalized nonzero bounded ancient solution of the centered self-similar Navier–Stokes equation. The construction keeps track of pressure gauges, local compactness, stability of the local energy inequality, and the velocity concentration supplied by local Caffarelli–Kohn–Nirenberg theory. It yields a smooth bounded ancient profile for the centered equation on $\mathbb{R}^3 \times \mathbb{R}$, together with a quantitative local L^3 nonvanishing condition.

We then organize the possible normalized Seregin ancient limits by standard analytic hypotheses: small amplitude, stationary L^3 , uniform L^3 -tightness, and structure or decay assumptions covered by cited Liouville theorems. The small-amplitude case is excluded by a bounded ancient perturbative Liouville theorem proved below, and the stationary L^3 case by the theorem of Nečas–Růžička–Šverák. The structure/decay class contains only hypotheses for which the cited Liouville theorems force $V \equiv 0$. The uniformly L^3 -tight class is closed in the companion tight-dynamics paper, and the remaining class is closed in the companion residual paper by local concentration analysis. Consequently the Type I exclusion obtained from this extraction is unconditional within the stated admissible finite-energy class.

1. INTRODUCTION AND MAIN RESULTS

1.1. Type I blow-up limits. Let $u : \mathbb{R}^3 \times (0, T) \rightarrow \mathbb{R}^3$ solve the incompressible Navier–Stokes equations

$$(1.1) \quad \partial_t u + (u \cdot \nabla)u + \nabla p = \Delta u, \quad \operatorname{div} u = 0.$$

The local regularity theory of Caffarelli–Kohn–Nirenberg [3] detects singular points through scale-invariant concentration of velocity and pressure. A terminal point $z_* = (x_*, T)$ is of Type I if

$$(1.2) \quad \limsup_{t \uparrow T} \sqrt{T-t} \|u(t)\|_{L^\infty(\mathbb{R}^3)} < \infty.$$

The blow-up extraction uses a selected velocity concentration sequence from the local concentration theorem and assumes only that this velocity concentration does not escape into the terminal slice after rescaling. The resulting admissibility condition is defined in [Theorem 2.4](#). It replaces a stronger all-scale compactness assumption on scale-invariant energy, pressure, and dissipation quantities by the non-escape condition needed for nontriviality of the ancient limit.

1.2. Normalized Seregin limits. For scales $\lambda \downarrow 0$, the parabolic rescaling about z_* is

$$u^{(\lambda)}(x, t) = \lambda u(x_* + \lambda x, T + \lambda^2 t), \quad p^{(\lambda)}(x, t) = \lambda^2 p(x_* + \lambda x, T + \lambda^2 t).$$

The rescaled time intervals exhaust $(-\infty, 0)$. The Type I bound becomes a uniform bound

$$\|u^{(\lambda)}(t)\|_{L^\infty(\mathbb{R}^3)} \leq M(-t)^{-1/2}$$

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on compact subintervals of $(-\infty, 0)$, and compactness produces an unrenormalized ancient solution (U, Q) . The associated centered variables are

$$y = \frac{x}{\sqrt{-t}}, \quad \tau = -\log(-t), \quad V(y, \tau) = \sqrt{-t} U(y\sqrt{-t}, t).$$

They transform the ancient Navier–Stokes solution into the centered equation

$$(1.3) \quad \partial_\tau V + (V \cdot \nabla) V + \nabla \Pi = \Delta V - \frac{1}{2}(V + y \cdot \nabla V), \quad \operatorname{div} V = 0.$$

The selected velocity concentration at the singular point persists through the extraction and gives a compact cylinder $K \Subset \mathbb{R}^3 \times \mathbb{R}$ and a constant $\eta_0 > 0$ such that

$$(1.4) \quad \iint_K |V|^3 dy d\tau \geq \eta_0.$$

1.3. Liouville alternatives and the remaining case. No unconditional Liouville theorem is known for all bounded ancient solutions of (1.3). We therefore work with a finite decomposition based on explicit Liouville inputs. A normalized Seregin limit is compared with the following analytic classes:

$$\mathcal{C}_{\text{small}}, \quad \mathcal{C}_{\text{stat}, L^3}, \quad \mathcal{C}_{L^3\text{-tight}}, \quad \mathcal{C}_{\text{known}}, \quad \mathcal{R}(\mathcal{S}).$$

They denote, respectively, small amplitude, stationary L^3 , uniform L^3 -tightness, structure or decay covered by cited Liouville theorems, and the complement of these four classes inside a fixed collection $\mathcal{S}$ of normalized limits. The last set represents no additional analytic mechanism; it is the remaining case after the stated Liouville alternatives have been removed.

1.4. Main theorems.

Theorem 1.1 (Normalized Type I blow-up limit). *Let (u, p, z_*) be an admissible finite-energy Type I singularity. Then there exist scales $\lambda_k \downarrow 0$, pressure gauges $a_k(t)$, an ancient suitable weak solution (U, Q) on $\mathbb{R}^3 \times (-\infty, 0)$, and a centered profile (V, Π) on $\mathbb{R}^3 \times \mathbb{R}$, such that, after passing to a subsequence,*

$$u_k \rightarrow U \quad \text{strongly in } L_{\text{loc}}^q(\mathbb{R}^3 \times (-\infty, 0)), \quad 1 \leq q < \infty,$$

and

$$p_k - a_k(t) \rightharpoonup Q \quad \text{weakly in } L_{\text{loc}}^{3/2}(\mathbb{R}^3 \times (-\infty, 0)).$$

The centered field V is smooth, bounded, solves (1.3), and satisfies the local L^3 nonvanishing condition (1.4).

For each $V \in \mathcal{S}$, the class definitions give the exhaustive alternative

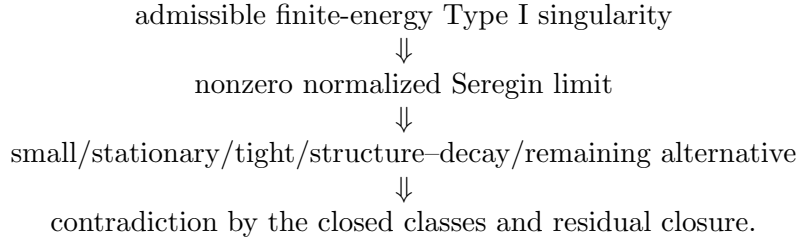
$$V \in \mathcal{C}_{\text{small}} \cup \mathcal{C}_{\text{stat}, L^3} \cup \mathcal{C}_{L^3\text{-tight}} \cup \mathcal{C}_{\text{known}} \cup \mathcal{R}(\mathcal{S}).$$

This is recorded later as [Theorem 6.9](#). It is a direct consequence of the definition of the remaining class, not an independent Liouville theorem.

Theorem 1.2 (Integrated Type I exclusion). *No admissible finite-energy Type I singularity exists.*

The proof uses two companion results. The uniformly L^3 -tight class is excluded in Paper II [6]. The remaining class is excluded in the residual closure paper [7]. The structure/decay class is restricted to hypotheses covered by cited Liouville theorems.

The dependency chain is



- Small amplitude: excluded here in [Section 7](#).
- Stationary L^3 : excluded here by [\[15\]](#).
- Uniformly L^3 -tight: proved in [Theorem 7.6](#), citing Paper II.
- Known structure/decay: excluded by the cited theorems in [Theorem 7.7](#).
- Remaining class: proved in [Theorem 8.1](#), citing the residual paper.

1.5. Dependencies and discharged inputs. The proof of the extraction theorem uses the local velocity concentration input stated as [Theorem 2.7](#); this is the only external result needed for the blow-up construction. The uniformly L^3 -tight Liouville theorem and the remaining-class assertion are not used to construct the ancient limit. They are invoked only after the class exhaustion, and both are proved in companion papers: [Theorem 7.6](#) is Paper II's endpoint L^3 closure, and [Theorem 8.1](#) is the residual closure theorem. The structure/decay class is restricted by definition to assumptions covered by the cited Liouville theorems. Thus the final Type I exclusion uses no assumption beyond the Paper 0 admissibility condition, the extraction proved here, and the closed companion inputs cited above.

1.6. Organization of the paper. [Section 2](#) records suitable solutions, scale-invariant quantities, the admissibility condition, scaling, pressure gauges, and local concentration. [Section 3](#) proves the compactness theorem for Type I rescalings away from the terminal time and constructs the ancient suitable weak limit. [Section 4](#) proves persistence of concentration and nontriviality. [Section 5](#) derives the centered equation and defines normalized Seregin limits and collections. [Section 6](#) defines the Liouville classes and the remaining class. [Section 7](#) records the class exclusions and the uniformly tight input. [Section 8](#) proves the remaining-case criterion. [Appendix A](#) records the elementary symmetries of the centered equation.

2. SUITABLE SOLUTIONS, SCALE-INVARIANT QUANTITIES, AND LOCAL CONCENTRATION

For $z_0 = (x_0, t_0)$, write

$$Q_r(z_0) = B_r(x_0) \times (t_0 - r^2, t_0).$$

When $z_0 = (0, 0)$, write $Q_r = Q_r(0, 0)$. The spatial average of f over $B_r(x_0)$ is

$$(f)_{B_r(x_0)}(t) = \frac{1}{|B_r|} \int_{B_r(x_0)} f(x, t) dx.$$

Throughout the paper, $\mathbb{P}$ denotes the Helmholtz–Leray projection onto divergence-free vector fields on $\mathbb{R}^3$.

Definition 2.1 (Terminal singular point). Let (u, p) be a suitable weak solution on $\mathbb{R}^3 \times (0, T)$. A terminal point $z_* = (x_*, T)$ is regular if there are $r > 0$ and $K < \infty$ such that

$$\operatorname{ess\,sup}_{Q_r(z_*)} |u| \leq K.$$

It is singular otherwise.

Definition 2.2 (Suitable weak solution). Let $\Omega \subset \mathbb{R}^3 \times \mathbb{R}$ be open. A pair (u, p) is a suitable weak solution of (1.1) in Ω if

$$u \in L_t^\infty L_x^2(\Omega') \cap L_t^2 \dot{H}_x^1(\Omega'), \quad p \in L^{3/2}(\Omega')$$

for every compact cylinder $\Omega' \Subset \Omega$, if (u, p) solves (1.1) distributionally, $\operatorname{div} u = 0$, and if the local energy inequality

$$\begin{aligned} & \int_{\mathbb{R}^3} |u(x, t_2)|^2 \phi(x, t_2) dx + 2 \int_{t_1}^{t_2} \int_{\mathbb{R}^3} |\nabla u|^2 \phi dx dt \\ & \leq \int_{\mathbb{R}^3} |u(x, t_1)|^2 \phi(x, t_1) dx + \int_{t_1}^{t_2} \int_{\mathbb{R}^3} |u|^2 (\partial_t \phi + \Delta \phi) dx dt \\ & + \int_{t_1}^{t_2} \int_{\mathbb{R}^3} (|u|^2 + 2p) u \cdot \nabla \phi dx dt \end{aligned} \quad (2.1)$$

holds for every non-negative $\phi \in C_c^\infty(\Omega)$ and almost every $t_1 < t_2$ for which the displayed terms are defined.

This is the Caffarelli–Kohn–Nirenberg notion of suitability [3]; the global finite-energy background is Leray’s Leray–Hopf class [11].

Definition 2.3 (Suitable Leray–Hopf solution on $\mathbb{R}^3 \times (0, T)$). A suitable Leray–Hopf solution on $\mathbb{R}^3 \times (0, T)$ is a suitable weak solution on $\mathbb{R}^3 \times (0, T)$ satisfying

$$u \in L^\infty(0, T; L^2(\mathbb{R}^3)) \cap L^2(0, T; \dot{H}^1(\mathbb{R}^3)), \quad u \in C_w([0, T]; L^2(\mathbb{R}^3)),$$

and the global Leray energy inequality. The pressure is taken in $L_{\text{loc}}^{3/2}(\mathbb{R}^3 \times (0, T))$, modulo functions of time.

Define

$$A(u; z_0, r) = r^{-1} \operatorname{ess\,sup}_{t_0 - r^2 < t < t_0} \int_{B_r(x_0)} |u(x, t)|^2 dx, \quad (2.2)$$

$$C(u; z_0, r) = r^{-2} \iint_{Q_r(z_0)} |u|^3 dx dt, \quad D(p; z_0, r) = r^{-2} \iint_{Q_r(z_0)} |p - (p)_{B_r(x_0)}(t)|^{3/2} dx dt, \quad (2.3)$$

$$E(u; z_0, r) = r^{-1} \iint_{Q_r(z_0)} |\nabla u|^2 dx dt, \quad (2.4)$$

Definition 2.4 (Admissible finite-energy Type I singularity). A triple (u, p, z_*) , with $z_* = (x_*, T)$, is an admissible finite-energy Type I singularity if:

- (i) (u, p) is a suitable Leray–Hopf solution on $\mathbb{R}^3 \times (0, T)$;
- (ii) $z_* = (x_*, T)$ is a singular point;
- (iii) the Type I bound (1.2) holds;
- (iv) there are scales $\lambda_k \downarrow 0$ and a constant $\eta_v > 0$ such that

$$C(u; z_*, \lambda_k) \geq \eta_v \quad \text{for every } k;$$

- (v) for the corresponding rescaled velocities

$$u_k(x, t) = \lambda_k u(x_* + \lambda_k x, T + \lambda_k^2 t),$$

the velocity no-escape condition holds:

$$\limsup_{\sigma \downarrow 0} \sup_k \int_{-\sigma}^0 \int_{B_1} |u_k(x, t)|^3 dx dt = 0. \quad (2.5)$$

The sequence $\{\lambda_k\}$ is called an admissible concentration sequence at z_* .

Remark 2.5. Item (iv) is the velocity-only concentration supplied by the local concentration theorem; see [Theorem 2.7](#). Thus, for a singular point, the additional admissibility requirement is the no-escape condition (2.5). This is the only terminal-slice compactness assumption used to prove that the ancient limit is nonzero. No uniform bound on $A + C + D + E$ over all terminal scales is assumed.

Lemma 2.6 (Uniform kinetic energy implies no escape). *Let (u, p) satisfy the Type I bound (1.2), and let $\lambda_k \downarrow 0$. Suppose that*

$$\sup_k A(u; z_*, \lambda_k) \leq B < \infty.$$

Then the corresponding rescaled velocities satisfy the no-escape condition (2.5).

Proof. Choose $M < \infty$ and $t_0 < T$ such that

$$\|u(t)\|_{L^\infty(\mathbb{R}^3)} \leq M(T - t)^{-1/2}, \quad t_0 < t < T.$$

For k sufficiently large, $T + \lambda_k^2 t > t_0$ whenever $-1 < t < 0$. Hence

$$\|u_k(t)\|_{L^\infty(B_1)} \leq M(-t)^{-1/2}, \quad -1 < t < 0.$$

By scale invariance of A ,

$$\operatorname{ess\,sup}_{-1 < t < 0} \int_{B_1} |u_k(x, t)|^2 dx = A(u; z_*, \lambda_k) \leq B.$$

Therefore, for $0 < \sigma < 1$,

$$\begin{aligned} \int_{-\sigma}^0 \int_{B_1} |u_k|^3 dx dt &\leq \int_{-\sigma}^0 \|u_k(t)\|_{L^\infty(B_1)} \int_{B_1} |u_k(x, t)|^2 dx dt \\ &\leq MB \int_{-\sigma}^0 (-t)^{-1/2} dt = 2MB\sigma^{1/2}. \end{aligned}$$

Taking the supremum over the tail $k \geq k_0$ and then letting $\sigma \downarrow 0$ gives (2.5) for the tail of the sequence. For each of the finitely many remaining indices $1 \leq k < k_0$, the rescaled velocity is defined on $B_1 \times (-\delta_k, 0)$ for some $\delta_k > 0$, and belongs to L^3 there. Therefore absolute continuity of the L^3 -integral gives

$$\lim_{\sigma \downarrow 0} \max_{1 \leq k < k_0} \int_{-\sigma}^0 \int_{B_1} |u_k|^3 dx dt = 0.$$

Combining the finite initial segment with the tail proves the full supremum in (2.5). $\square$

Proposition 2.7 (Local velocity concentration). *Let (u, p) be suitable in a parabolic neighborhood of $z_* = (x_*, T)$. If z_* is singular, then there exist $\lambda_k \downarrow 0$ and $\eta_v > 0$ such that*

$$C(u; z_*, \lambda_k) \geq \eta_v \quad \text{for every } k.$$

Proof. This is the velocity-only concentration corollary of the local concentration result in Paper 0 [16]. That result follows from the velocity ε -regularity criterion: if

$$\limsup_{r \downarrow 0} C(u; z_*, r) = 0,$$

then u is bounded in a backward parabolic cylinder ending at z_* , contradicting singularity of z_* . $\square$

Lemma 2.8 (Scaling). *Let (u, p) be suitable in a neighborhood of (x_*, T) . For $\lambda > 0$, set*

$$u^\lambda(x, t) = \lambda u(x_* + \lambda x, T + \lambda^2 t), \quad p^\lambda(x, t) = \lambda^2 p(x_* + \lambda x, T + \lambda^2 t).$$

Then (u^λ, p^λ) is suitable on the rescaled domain. Moreover A, C, D, E are invariant under this scaling.

Proof. Suitability of the rescaled pair and preservation of pressure gauges follow by applying (2.1) to the test function

$$\phi^\lambda(X, S) = \phi\left(\frac{X - x_*}{\lambda}, \frac{S - T}{\lambda^2}\right)$$

and then changing variables $(X, S) = (x_* + \lambda x, T + \lambda^2 t)$. The differential identities

$$\begin{aligned} \partial_t u^\lambda &= \lambda^3 (\partial_t u)(X, S), & (u^\lambda \cdot \nabla) u^\lambda &= \lambda^3 ((u \cdot \nabla) u)(X, S), \\ \nabla p^\lambda &= \lambda^3 (\nabla p)(X, S), & \Delta u^\lambda &= \lambda^3 (\Delta u)(X, S) \end{aligned}$$

give the rescaled equation and the divergence condition.

For the pressure average,

$$(p^\lambda)_{B_r}(t) = \lambda^2 (p)_{B_{\lambda r}(x_*)}(S).$$

Consequently

$$\iint_{Q_r} |u^\lambda|^3 dx dt = \lambda^{-2} \iint_{Q_{\lambda r}(z_*)} |u|^3 dX dS$$

and

$$\iint_{Q_r} |p^\lambda - (p^\lambda)_{B_r}(t)|^{3/2} dx dt = \lambda^{-2} \iint_{Q_{\lambda r}(z_*)} |p - (p)_{B_{\lambda r}(x_*)}(S)|^{3/2} dX dS.$$

Multiplication by r^{-2} therefore gives invariance of C and D . The remaining two quantities have the same critical dimensional balance:

$$\begin{aligned} \int_{B_r} |u^\lambda(x, t)|^2 dx &= \lambda^{-1} \int_{B_{\lambda r}(x_*)} |u(X, S)|^2 dX, \\ \iint_{Q_r} |\nabla u^\lambda|^2 dx dt &= \lambda^{-1} \iint_{Q_{\lambda r}(z_*)} |\nabla u|^2 dX dS. \end{aligned}$$

Multiplication by r^{-1} on the rescaled cylinder is the same as multiplication by $(\lambda r)^{-1}$ on the original cylinder. Hence A and E are invariant as well. $\square$

Lemma 2.9 (Pressure gauges do not change the equations). *Let (v, q) be suitable in a cylinder Q , and let $\alpha(t) \in L_{\text{loc}}^{3/2}$ depend only on time. Then $(v, q - \alpha(t))$ satisfies the same distributional Navier–Stokes equation, the same divergence condition, and the same local energy inequality.*

Proof. In the distributional equation, replacing q by $q - \alpha(t)$ adds $-\nabla \alpha(t) = 0$. In the local energy inequality the only new term is

$$-2 \int \alpha(t) v(x, t) \cdot \nabla \phi(x, t) dx dt.$$

The product is integrable on the support of ϕ : by suitability, $v \in L_t^\infty L_x^2$ locally, and

$$\left| \int v(\cdot, t) \cdot \nabla \phi(\cdot, t) dx \right| \leq \|v(t)\|_{L^2(\text{supp } \phi(t))} \|\nabla \phi(t)\|_{L^2}.$$

For almost every t , $\text{div } v(\cdot, t) = 0$ in distributions and $\phi(\cdot, t)$ is compactly supported, so

$$\int v(x, t) \cdot \nabla \phi(x, t) dx = 0.$$

Thus the additional term vanishes. $\square$

3. THE TYPE I BLOW-UP SEQUENCE AND COMPACTNESS

3.1. The rescaled sequence. Choose $M < \infty$ so that, after decreasing $T_0 < T$ if necessary,

$$(3.1) \quad \|u(t)\|_{L^\infty(\mathbb{R}^3)} \leq \frac{M}{\sqrt{T-t}}, \quad T_0 < t < T.$$

For $\lambda_k \downarrow 0$, define

$$(3.2) \quad u_k(x, t) = \lambda_k u(x_* + \lambda_k x, T + \lambda_k^2 t), \quad p_k(x, t) = \lambda_k^2 p(x_* + \lambda_k x, T + \lambda_k^2 t).$$

Then u_k is defined on $\mathbb{R}^3 \times (-T/\lambda_k^2, 0)$, and these domains exhaust $\mathbb{R}^3 \times (-\infty, 0)$.

3.2. Inherited Type I bounds.

Lemma 3.1 (Inherited Type I bound). *For every compact interval $I \Subset (-\infty, 0)$,*

$$\sup_k \|u_k\|_{L^\infty(\mathbb{R}^3 \times I)} < \infty.$$

More precisely, for every fixed $t < 0$ and all sufficiently large k ,

$$\|u_k(t)\|_{L^\infty(\mathbb{R}^3)} \leq \frac{M}{\sqrt{-t}}.$$

Proof. Since $T - (T + \lambda_k^2 t) = \lambda_k^2(-t)$, (3.1) gives, for all large k ,

$$\|u_k(t)\|_\infty = \lambda_k \|u(T + \lambda_k^2 t)\|_\infty \leq \lambda_k \frac{M}{\sqrt{\lambda_k^2(-t)}} = \frac{M}{\sqrt{-t}}.$$

Taking the supremum over $t \in I$ gives the interval statement. $\square$

3.3. Local energy, pressure, and time-derivative estimates.

Lemma 3.2 (Pressure gauges and local pressure bounds). *After fixing the whole-space Leray pressure representative below, set*

$$a_k(t) = (p_k)_{B_1}(t)$$

whenever the rescaled solution is defined. Then for every

$$Q = B_R \times [-S, -\sigma] \Subset \mathbb{R}^3 \times (-\infty, 0)$$

one has

$$\sup_k \|p_k - a_k(t)\|_{L^{3/2}(Q)} < \infty.$$

Proof. For large k , the original times $T + \lambda_k^2 t$, $t \in [-S, -\sigma]$, lie in the interval where (3.1) holds. On those slices $u(t) \in L^2(\mathbb{R}^3) \cap L^\infty(\mathbb{R}^3)$, hence

$$u_i(t)u_j(t) \in L^1(\mathbb{R}^3) \cap L^{3/2}(\mathbb{R}^3).$$

Use the whole-space Leray pressure representative

$$p^L = \mathcal{R}_i \mathcal{R}_j (u_i u_j),$$

modulo functions of time. We use the standard convention

$$\widehat{\mathcal{R}_i f}(\xi) = i\xi_i |\xi|^{-1} \widehat{f}(\xi),$$

so that $\mathcal{R}_i \mathcal{R}_j$ has multiplier $-\xi_i \xi_j |\xi|^{-2}$ and $-\Delta(\mathcal{R}_i \mathcal{R}_j f) = \partial_i \partial_j f$. Thus this is the pressure reconstruction from $-\Delta p = \partial_i \partial_j (u_i u_j)$ and the Calderon-Zygmund theory of Riesz transforms [21, 20]. Since $u(t) \in L^2(\mathbb{R}^3) \cap L^\infty(\mathbb{R}^3)$, interpolation gives $u(t) \in L^3(\mathbb{R}^3)$, hence $u_i(t)u_j(t) \in L^{3/2}(\mathbb{R}^3)$ and $p^L(t) \in L^{3/2}(\mathbb{R}^3)$ for a.e. t .

The given pressure and p^L differ by a function of time. Indeed, applying the Helmholtz projection to the whole-space weak momentum equation gives

$$\partial_t u - \Delta u + \mathbb{P} \operatorname{div}(u \otimes u) = 0$$

in solenoidal distributions. The complementary gradient part of $\operatorname{div}(u \otimes u)$ is precisely $-\nabla p^L$ with the convention above. Therefore the same velocity field solves the distributional equation with p^L in place of p . Subtracting the two formulations gives

$$\nabla(p - p^L) = 0 \quad \text{in } \mathcal{D}'(\mathbb{R}^3)$$

for a.e. time. By the de Rham theorem on $\mathbb{R}^3$, $p - p^L$ is spatially constant for a.e. time. Replacing the pressure by p^L is therefore a time-dependent gauge change and does not affect suitability by [Theorem 2.9](#).

Fix $\chi_R \in C_c^\infty(B_{8R})$ with $\chi_R = 1$ on B_{4R} . For $t \in [-S, -\sigma]$, choose the representative so that, in B_{2R} ,

$$p_k = q_k + h_k, \quad q_k = \mathcal{R}_i \mathcal{R}_j (\chi_R u_{k,i} u_{k,j}),$$

and $h_k(\cdot, t)$ is harmonic in B_{4R} for a.e. t . This choice is made before $a_k(t) = (p_k)_{B_1}(t)$ is evaluated. The harmonicity holds because $(1 - \chi_R)u_{k,i}u_{k,j}$ is supported outside B_{4R} , so the localized Riesz potential generated by this exterior part has zero Laplacian in B_{4R} . Calderon–Zygmund boundedness gives, for a.e. t ,

$$\|q_k(t)\|_{L^{3/2}(B_{2R})} \leq C \|u_k(t)\|_{L^3(B_{8R})}^2 \leq C(R, \sigma, M).$$

Integrating over $[-S, -\sigma]$ gives a uniform $L_{x,t}^{3/2}$ -bound for q_k . Let $c_{k,R}(t) = h_k(0, t)$, where $h_k(\cdot, t)$ is represented by its smooth harmonic representative. For $x \in B_R$, the kernel estimate

$$|K_{ij}(x - z) - K_{ij}(-z)| \leq CR|z|^{-4}, \quad |z| > 4R,$$

implies

$$|h_k(x, t) - c_{k,R}(t)| \leq CR \sum_{m=2}^{\infty} (2^m R)^{-4} \int_{2^m R < |z| < 2^{m+1} R} |u_k(z, t)|^2 dz.$$

Using [Theorem 3.1](#),

$$\int_{2^m R < |z| < 2^{m+1} R} |u_k(z, t)|^2 dz \leq CM^2 \sigma^{-1} (2^m R)^3,$$

and therefore

$$|h_k(x, t) - c_{k,R}(t)| \leq CM^2 \sigma^{-1} \sum_{m=2}^{\infty} 2^{-m}, \quad x \in B_R,$$

which implies

$$\|h_k(t) - c_{k,R}(t)\|_{L^\infty(B_R)} \leq C(M, \sigma).$$

Thus

$$\sup_k \|p_k - c_{k,R}(t)\|_{L^{3/2}(B_R \times [-S, -\sigma])} < \infty.$$

If $R < 1$, enlarge R to 1. Otherwise $B_1 \subset B_R$. Jensen's inequality gives, for a.e. t ,

$$|a_k(t) - c_{k,R}(t)|^{3/2} \leq |B_1|^{-1} \int_{B_1} |p_k - c_{k,R}(t)|^{3/2} dx.$$

Integrating in time and adding constants on B_R gives the desired bound for $p_k - a_k(t)$. $\square$

Lemma 3.3 (Compatibility of pressure gauges). *Let $B_1 \subset B_R$, $I \Subset (-\infty, 0)$, and suppose*

$$p_k - c_k(t) \text{ is bounded in } L^{3/2}(B_R \times I)$$

for some time-dependent gauges $c_k(t)$. If $a_k(t) = (p_k)_{B_1}(t)$, then $p_k - a_k(t)$ is bounded in $L^{3/2}(B_{R'} \times I)$ for every $R' \leq R$. Changing between such gauges changes the pressure only by a function of time and therefore does not affect the equation, the local energy inequality, or the centered equation.

Proof. For a.e. t , Jensen's inequality gives

$$|a_k(t) - c_k(t)|^{3/2} \leq |B_1|^{-1} \int_{B_1} |p_k - c_k(t)|^{3/2} dx.$$

Integrating in t and using $B_{R'} \subset B_R$ yields

$$\|a_k - c_k\|_{L^{3/2}(I)}^{3/2} \leq |B_1|^{-1} \|p_k - c_k(t)\|_{L^{3/2}(B_R \times I)}^{3/2}.$$

Thus

$$\|p_k - a_k(t)\|_{L^{3/2}(B_{R'} \times I)} \leq \|p_k - c_k(t)\|_{L^{3/2}(B_{R'} \times I)} + |B_{R'}|^{2/3} \|a_k - c_k\|_{L^{3/2}(I)}$$

and the right-hand side is bounded by

$$C(R') \|p_k - c_k(t)\|_{L^{3/2}(B_R \times I)}.$$

This proves the asserted bound. The last assertion is [Theorem 2.9](#), and centering transforms the time-dependent gauge into another time-dependent gauge. $\square$

Lemma 3.4 (Local energy and dissipation). *Let*

$$Q = B_R \times [-S, -\sigma] \Subset \mathbb{R}^3 \times (-\infty, 0).$$

Then

$$\sup_k \|u_k\|_{L_t^\infty L_x^2(Q)} + \sup_k \|\nabla u_k\|_{L^2(Q)} < \infty.$$

Proof. The $L_t^\infty L_x^2$ estimate follows from

$$\|u_k(t)\|_{L^2(B_R)} \leq |B_R|^{1/2} M \sigma^{-1/2}, \quad t \in [-S, -\sigma].$$

Choose $\phi \in C_c^\infty(B_{2R} \times [-2S, -\sigma/2])$, $0 \leq \phi \leq 1$, with $\phi = 1$ on Q . Apply the local energy inequality to $(u_k, p_k - a_k(t))$, using [Theorem 3.2](#) on the larger cylinder. For a.e. admissible times and then by monotone approximation of temporal cutoffs,

$$2 \iint |\nabla u_k|^2 \phi \leq \iint |u_k|^2 |\partial_t \phi + \Delta \phi| + \iint (|u_k|^2 + 2|p_k - a_k(t)|) |u_k| |\nabla \phi|.$$

The first term is bounded because $\partial_t \phi + \Delta \phi$ is bounded and u_k is uniformly bounded in L^∞ on the finite-measure support of ϕ . The cubic velocity term satisfies

$$\iint |u_k|^3 |\nabla \phi| \leq \|\nabla \phi\|_\infty \text{supp } \nabla \phi \|u_k\|_{L^\infty(\text{supp } \nabla \phi)}^3 \leq C(Q, \phi, M, \sigma).$$

For the pressure term, Hölder's inequality on $\text{supp } \nabla \phi$ gives

$$\iint |p_k - a_k(t)| |u_k| |\nabla \phi| \leq \|p_k - a_k(t)\|_{L^{3/2}} \|u_k \nabla \phi\|_{L^3},$$

and both factors are uniformly bounded by [Theorem 3.2](#) and [Theorem 3.1](#). Since $\phi = 1$ on Q , the dissipation bound follows. $\square$

Lemma 3.5 (Time derivative bound). *For every compact cylinder $Q \Subset \mathbb{R}^3 \times (-\infty, 0)$,*

$$\partial_t u_k \text{ is bounded in } L_t^{3/2} W_x^{-1, 3/2}(Q).$$

Proof. The equation is

$$\partial_t u_k = \Delta u_k - \operatorname{div}(u_k \otimes u_k) - \nabla(p_k - a_k(t)).$$

Let $Q = B_R \times [-S, -\sigma]$. The term Δu_k is bounded in $L_t^2 H_x^{-1}(Q)$ by [Theorem 3.4](#); since $W_0^{1,3}(B_R) \subset H_0^1(B_R)$ and the time interval is finite, this is also an $L_t^{3/2} W_x^{-1,3/2}$ -bound. Explicitly, for $\zeta \in W_0^{1,3}(B_R)$,

$$|\langle \Delta u_k, \zeta \rangle| = \left| \int_{B_R} \nabla u_k : \nabla \zeta \, dx \right| \leq C(R) \|\nabla u_k\|_{L^2(B_R)} \|\zeta\|_{W^{1,3}(B_R)}.$$

The L_t^2 -bound on ∇u_k , together with the finite length of $[-S, -\sigma]$, gives the claimed $L_t^{3/2}$ -bound.

For the nonlinear term, for $\zeta \in W_0^{1,3}(B_R)$,

$$\left| \int_{B_R} (u_k \otimes u_k) : \nabla \zeta \, dx \right| \leq \|u_k \otimes u_k\|_{L^{3/2}(B_R)} \|\nabla \zeta\|_{L^3(B_R)}.$$

Moreover

$$\|u_k \otimes u_k\|_{L^{3/2}(Q)} \leq \|u_k\|_{L^\infty(Q)} \|u_k\|_{L^{3/2}(Q)} \leq C(Q).$$

Thus $\operatorname{div}(u_k \otimes u_k)$ is bounded in $L_t^{3/2} W_x^{-1,3/2}(Q)$. For the pressure term,

$$\left| \int_{B_R} (p_k - a_k(t)) \operatorname{div} \zeta \, dx \right| \leq \|p_k - a_k(t)\|_{L^{3/2}(B_R)} \|\nabla \zeta\|_{L^3(B_R)}.$$

[Theorem 3.2](#) gives

$$p_k - a_k(t) \text{ bounded in } L^{3/2}(Q).$$

Hence $\nabla(p_k - a_k(t))$ is bounded in $L_t^{3/2} W_x^{-1,3/2}(Q)$. Summing the three terms gives the result. $\square$

3.4. Compactness away from the terminal time.

Proposition 3.6 (Compactness). *After passing to a subsequence,*

$$u_k \rightarrow U$$

strongly in $L_{\text{loc}}^q(\mathbb{R}^3 \times (-\infty, 0))$ for every finite q , and

$$p_k - a_k(t) \rightharpoonup Q \text{ weakly in } L_{\text{loc}}^{3/2}(\mathbb{R}^3 \times (-\infty, 0)).$$

Proof. Fix $Q_0 \Subset \mathbb{R}^3 \times (-\infty, 0)$. By [Theorem 3.4](#), u_k is bounded in $L_t^2 H_x^1(Q_0)$, and by [Theorem 3.5](#), $\partial_t u_k$ is bounded in $L_t^{3/2} W_x^{-1,3/2}(Q_0)$. On bounded balls,

$$H^1 \Subset L^2 \hookrightarrow W^{-1,3/2}.$$

The Aubin–Lions–Simon theorem [[2](#), [12](#), [19](#)] gives relative compactness in $L^2(Q_0)$. Hence, after passing to a subsequence, $u_k \rightarrow U$ strongly in $L^2(Q_0)$ and a.e. on Q_0 .

The inherited Type I bound gives a common $L^\infty(Q_0)$ bound, and the limit belongs to the same L^∞ ball by a.e. convergence. For $q \geq 2$,

$$\|u_k - U\|_{L^q(Q_0)}^q \leq (2 \sup_j \|u_j\|_{L^\infty(Q_0)} + 2\|U\|_{L^\infty(Q_0)})^{q-2} \|u_k - U\|_{L^2(Q_0)}^2.$$

For $1 \leq q < 2$, Hölder's inequality on the finite-measure set Q_0 gives

$$\|u_k - U\|_{L^q(Q_0)} \leq |Q_0|^{1/q-1/2} \|u_k - U\|_{L^2(Q_0)} \rightarrow 0.$$

Choose an exhaustion $Q^{(m)} \Subset \mathbb{R}^3 \times (-\infty, 0)$ by compact cylinders. The preceding argument gives a subsequence converging on $Q^{(1)}$; extracting successively on $Q^{(m)}$ and taking the diagonal subsequence gives strong local convergence on every compact cylinder.

For pressure, [Theorem 3.2](#) gives boundedness of $p_k - a_k(t)$ in $L^{3/2}(Q^{(m)})$ for every m . Since $L^{3/2}$ is reflexive, each restriction has a weakly convergent subsequence. The same diagonal extraction gives a single limit Q in $L_{\text{loc}}^{3/2}$, because the weak limits agree on overlaps by uniqueness of distributional limits. $\square$

3.5. The ancient suitable weak limit.

Lemma 3.7 (Limit equation). *The limit (U, Q) solves*

$$\partial_t U + (U \cdot \nabla)U + \nabla Q = \Delta U, \quad \operatorname{div} U = 0$$

in distributions on $\mathbb{R}^3 \times (-\infty, 0)$.

Proof. For $\varphi \in C_c^\infty(\mathbb{R}^3 \times (-\infty, 0); \mathbb{R}^3)$, the weak formulation for $(u_k, p_k - a_k(t))$ is

$$\iint [-u_k \cdot \partial_t \varphi - (u_k \otimes u_k) : \nabla \varphi + (p_k - a_k(t)) \operatorname{div} \varphi + \nabla u_k : \nabla \varphi] dx dt = 0.$$

The linear time term passes by strong L_{loc}^2 -convergence of u_k . The pressure term passes by weak $L_{\text{loc}}^{3/2}$ -convergence of $p_k - a_k(t)$, since $\operatorname{div} \varphi \in L^3$. The viscous term passes by weak L_{loc}^2 -convergence of ∇u_k , which follows from the uniform local dissipation bound. The nonlinear term passes because

$$\|u_k \otimes u_k - U \otimes U\|_{L^1(\operatorname{supp} \varphi)} \leq \|u_k - U\|_{L^2} (\|u_k\|_{L^2} + \|U\|_{L^2}) \rightarrow 0.$$

Testing with gradients of scalar functions gives $\operatorname{div} U = 0$. $\square$

Lemma 3.8 (Stability of suitability). *The pair (U, Q) is a suitable weak solution on $\mathbb{R}^3 \times (-\infty, 0)$.*

Proof. Use the distributional form of the local energy inequality. For every non-negative $\phi \in C_c^\infty(\mathbb{R}^3 \times (-\infty, 0))$,

$$2 \iint |\nabla u_k|^2 \phi \leq \iint |u_k|^2 (\partial_t \phi + \Delta \phi) + \iint (|u_k|^2 + 2P_k) u_k \cdot \nabla \phi, \quad P_k = p_k - a_k(t).$$

This follows from (2.1) by applying (2.1) with temporal cutoffs which increase to 1 on the time projection of $\operatorname{supp} \phi$. The monotone convergence theorem passes the nonnegative dissipation term to the displayed distributional form.

The terms with $|u_k|^2$ converge strongly in L^1 because

$$\||u_k|^2 - |U|^2\|_{L^1(\operatorname{supp} \phi)} \leq \|u_k - U\|_{L^2} (\|u_k\|_{L^2} + \|U\|_{L^2}) \rightarrow 0.$$

The cubic flux converges strongly since $u_k \rightarrow U$ strongly in L_{loc}^3 , hence

$$|u_k|^2 u_k \rightarrow |U|^2 U \quad \text{in } L_{\text{loc}}^1.$$

For the pressure flux, $u_k \cdot \nabla \phi \rightarrow U \cdot \nabla \phi$ strongly in L^3 , while $P_k \rightharpoonup Q$ weakly in $L^{3/2}$. Therefore

$$\iint P_k u_k \cdot \nabla \phi \rightarrow \iint Q U \cdot \nabla \phi.$$

Finally, $\nabla u_k \rightharpoonup \nabla U$ in L_{loc}^2 , and

$$w \mapsto \iint |w|^2 \phi$$

is weakly lower semicontinuous because $\phi \geq 0$ and the expression equals $\|\sqrt{\phi}w\|_{L^2}^2$. Passing to the limit gives the local energy inequality for (U, Q) in distributional form. The endpoint form in [Theorem 2.2](#) is recovered by applying the distributional inequality to standard time cutoffs approximating characteristic functions of (t_1, t_2) ; weak L^2_{loc} -continuity of local energy-class solutions gives the endpoint traces for a.e. $t_1 < t_2$. $\square$

Lemma 3.9 (Inherited ancient Type I bound). *For every compact interval $I = [-S, -\sigma] \Subset (-\infty, 0)$,*

$$\|U\|_{L^\infty(\mathbb{R}^3 \times I)} \leq M\sigma^{-1/2}.$$

Proof. [Theorem 3.1](#) gives

$$\|u_k\|_{L^\infty(\mathbb{R}^3 \times I)} \leq M\sigma^{-1/2}.$$

Fix $R < \infty$. Since $u_k \rightarrow U$ a.e. on $B_R \times I$, the essential supremum of U on $B_R \times I$ cannot exceed $M\sigma^{-1/2}$. If it did, $|u_k|$ would exceed this bound by a positive amount on a positive measure subset for large k . Letting $R \rightarrow \infty$ through integers gives the result on $\mathbb{R}^3 \times I$. $\square$

Proposition 3.10 (Ancient suitable Type I limit). *The pair (U, Q) is an ancient suitable weak solution on $\mathbb{R}^3 \times (-\infty, 0)$ satisfying the interval Type I bound in [Theorem 3.9](#).*

Proof. Combine [Theorems 3.7](#) to [3.9](#). $\square$

4. PERSISTENCE OF CONCENTRATION AND NONTRIVIALITY

4.1. Persistence of terminal concentration away from the terminal slice. The singularity enters the proof through the velocity concentration sequence in [Theorem 2.4](#), supplied by [Theorem 2.7](#). The no-escape condition says that this velocity mass does not concentrate entirely in the terminal layer $t = 0$ after rescaling.

4.2. Scale selection.

Lemma 4.1 (Scale selection from local concentration). *Let $\lambda_k \downarrow 0$ be an admissible concentration sequence in the sense of [Theorem 2.4](#). Then there are $\sigma \in (0, 1)$, a compact cylinder*

$$K_0 = B_1 \times (-1, -\sigma) \Subset \mathbb{R}^3 \times (-\infty, 0),$$

and $\eta_1 > 0$ such that

$$(4.1) \quad \iint_{K_0} |u_k|^3 dx dt \geq \eta_1$$

for all k .

Proof. By scale invariance of C ,

$$\int_{-1}^0 \int_{B_1} |u_k|^3 dx dt = C(u; z_*, \lambda_k) \geq \eta_v.$$

The no-escape condition [\(2.5\)](#) gives $\sigma \in (0, 1)$ such that

$$\sup_k \int_{-\sigma}^0 \int_{B_1} |u_k|^3 dx dt \leq \frac{\eta_v}{2}.$$

Therefore

$$\int_{-1}^{-\sigma} \int_{B_1} |u_k|^3 dx dt \geq \frac{\eta_v}{2}$$

for every k . The assertion follows with $\eta_1 = \eta_v/2$. $\square$

4.3. Nonzero ancient limits.

Proposition 4.2 (Nonzero ancient limit). *For the scales chosen in [Theorem 4.1](#), the limit U in [Theorem 3.10](#) is nonzero. More precisely, there are a compact cylinder $K_0 \Subset \mathbb{R}^3 \times (-\infty, 0)$ and $\eta_1 > 0$ such that*

$$\iint_{K_0} |U|^3 dx dt \geq \eta_1.$$

Proof. Let $K_0 = B_1 \times (-1, -\sigma)$ and $\eta_1 > 0$ be given by [Theorem 4.1](#). Since $K_0 \Subset \mathbb{R}^3 \times (-\infty, 0)$, the compactness theorem gives, after passing to the subsequence used to define U ,

$$u_k \rightarrow U \quad \text{strongly in } L^3(K_0).$$

Hence

$$\int_{K_0} |U|^3 dx dt = \lim_{k \rightarrow \infty} \int_{K_0} |u_k|^3 dx dt \geq \eta_1.$$

□

The preceding proposition is the point at which singularity enters the compactness argument. Without the velocity concentration and the no-escape condition, the compactness theorem alone could produce the zero ancient solution.

5. CENTERED VARIABLES AND NORMALIZED SEREGIN LIMITS

5.1. The centered self-similar equation. Set

$$y = \frac{x}{\sqrt{-t}}, \quad \tau = -\log(-t), \quad V(y, \tau) = \sqrt{-t} U(y\sqrt{-t}, t),$$

and

$$\Pi(y, \tau) = (-t)Q(y\sqrt{-t}, t),$$

up to addition of a function of τ .

Lemma 5.1 (Renormalized equation). *The pair (V, Π) solves [\(1.3\)](#).*

Proof. Write $s = -t = e^{-\tau}$, $x = s^{1/2}y$, and

$$U(x, t) = s^{-1/2}V(y, \tau).$$

The identities are first obtained in distributions by testing the equation for U against the inverse change of variables. After [Theorem 5.2](#), the same identities hold classically for the smooth representative. Because $s_t = -1$, $\tau_t = s^{-1}$, and $y_t = (2s)^{-1}y$,

$$\partial_t U = s^{-3/2} \left(\partial_\tau V + \frac{1}{2}V + \frac{1}{2}y \cdot \nabla V \right),$$

$$(U \cdot \nabla_x)U = s^{-3/2}(V \cdot \nabla)V, \quad \Delta_x U = s^{-3/2}\Delta V, \quad \nabla_x Q = s^{-3/2}\nabla \Pi.$$

Substitution into the equation for U gives [\(1.3\)](#), and the divergence condition is preserved by the spatial scaling. □

5.2. Boundedness, smoothness, and local energy.

Lemma 5.2 (Smoothness). *The ancient solution U is smooth on $\mathbb{R}^3 \times (-\infty, 0)$, and (V, Π) is smooth on $\mathbb{R}^3 \times \mathbb{R}$ after fixing a pressure gauge.*

Proof. On each compact cylinder $Q_0 \Subset \mathbb{R}^3 \times (-\infty, 0)$, [Theorem 3.9](#) gives $U \in L^\infty(Q_0)$. Hence $U \in L_t^s L_x^r(Q_0)$ for every finite r, s . Choose $r = s = 10$, so

$$\frac{2}{s} + \frac{3}{r} = \frac{1}{2} < 1.$$

Serrin's interior regularity criterion [\[18, 20\]](#) applies and gives local Hölder regularity of U on a slightly smaller cylinder. Since Q_0 was arbitrary, this removes every possible interior singular point in $\mathbb{R}^3 \times (-\infty, 0)$. The bootstrap starts from

$$\partial_t U - \Delta U + \nabla Q = -\operatorname{div}(U \otimes U), \quad \operatorname{div} U = 0,$$

with $U \otimes U$ locally bounded after Serrin regularity. Interior Stokes estimates on nested cylinders give $W_q^{2,1}$ -regularity for every finite q . Parabolic Sobolev embedding then improves the Hölder regularity of ∇U , and differentiating the equation and iterating gives

$$U \in C^\infty(\mathbb{R}^3 \times (-\infty, 0)).$$

The pressure is recovered locally from $-\Delta Q = \partial_i \partial_j (U_i U_j)$ plus a harmonic function and is smooth after fixing a time-dependent gauge. The centered change of variables is a smooth diffeomorphism for $t < 0$, so V and Π are smooth. $\square$

Corollary 5.3 (Classical Type I bound). *For the smooth representative from [Theorem 5.2](#),*

$$\|U(t)\|_{L^\infty(\mathbb{R}^3)} \leq \frac{M}{\sqrt{-t}}, \quad t < 0.$$

Proof. Fix $t_0 < 0$. Choose $\delta_n \downarrow 0$ with

$$I_n = [t_0 - \delta_n, t_0 + \delta_n] \Subset (-\infty, 0).$$

[Theorem 3.9](#) gives

$$\|U\|_{L^\infty(\mathbb{R}^3 \times I_n)} \leq M(-t_0 - \delta_n)^{-1/2}.$$

Since the representative is continuous, the essential spacetime bound holds pointwise. Letting $n \rightarrow \infty$ gives the estimate at t_0 . $\square$

Lemma 5.4 (Renormalized boundedness and nontriviality). *The centered field satisfies*

$$\|V\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})} \leq M.$$

Moreover [\(1.4\)](#) holds for some compact cylinder $K \Subset \mathbb{R}^3 \times \mathbb{R}$ and $\eta_0 > 0$.

Proof. The L^∞ bound follows from [Theorem 5.3](#):

$$|V(y, \tau)| = \sqrt{-t} |U(y\sqrt{-t}, t)| \leq M.$$

For nontriviality, [Theorem 4.2](#) gives a positive local L^3 lower bound on a compact cylinder in (x, t) . Under $(x, t) = (e^{-\tau/2}y, -e^{-\tau})$,

$$dx dt = e^{-5\tau/2} dy d\tau, \quad U = e^{\tau/2} V,$$

and hence

$$|U|^3 dx dt = e^{-\tau} |V|^3 dy d\tau.$$

On the compact image cylinder the factor $e^{-\tau}$ is bounded above and below by positive constants, so the positive lower bound becomes [\(1.4\)](#), with a possibly different η_0 . $\square$

Lemma 5.5 (Local energy bounds in centered variables). *Let V be obtained from an admissible finite-energy Type I singularity. For every $R < \infty$ and compact interval $I \subset \mathbb{R}$,*

$$\int_I \int_{B_R} |V|^2 dy d\tau + \int_I \int_{B_R} |\nabla V|^2 dy d\tau < \infty.$$

After subtracting a function of τ , the associated pressure satisfies

$$\Pi - b(\tau) \in L^{3/2}(B_R \times I).$$

Proof. The unrenormalized limit (U, Q) is suitable and has local energy, dissipation, and pressure bounds on compact negative-time cylinders by [Theorems 3.2, 3.4](#) and [3.10](#); the bounds for the limit follow from the weak lower semicontinuity and weak compactness already used in [Theorem 3.8](#). On each compact (y, τ) -cylinder, the map

$$(y, \tau) \mapsto (x, t) = (e^{-\tau/2}y, -e^{-\tau})$$

has Jacobian $dx dt = e^{-5\tau/2} dy d\tau$, with

$$V = e^{-\tau/2}U, \quad \nabla_y V = e^{-\tau} \nabla_x U, \quad \Pi = e^{-\tau}Q.$$

If $I = [\tau_1, \tau_2]$, then $e^{-\tau}$ is bounded above and below by positive constants on I . Therefore

$$\int_I \int_{B_R} |V|^2 dy d\tau \leq C(I) \int_{-e^{-\tau_1}}^{-e^{-\tau_2}} \int_{B_{C(I)R}} |U|^2 dx dt,$$

and similarly

$$\int_I \int_{B_R} |\nabla_y V|^2 dy d\tau \leq C(I) \int_{-e^{-\tau_1}}^{-e^{-\tau_2}} \int_{B_{C(I)R}} |\nabla_x U|^2 dx dt.$$

For the pressure, subtract a spatial-average gauge $a(t)$ for Q on the corresponding x -ball and set $b(\tau) = e^{-\tau}a(-e^{-\tau})$. Then $\Pi - b = e^{-\tau}(Q - a)$, and the same bounded-weight change of variables gives $\Pi - b \in L^{3/2}(B_R \times I)$. $\square$

Remark 5.6. [Theorem 5.5](#) is local. It does not give uniform L^3 -tightness in y , nor a global $L_\tau^\infty L_y^3$ bound. This is the reason for retaining the remaining case.

5.3. Normalized Seregin ancient limits.

Definition 5.7 (Normalized Seregin ancient limit). A smooth bounded solution (V, Π) of [\(1.3\)](#) is a normalized Seregin ancient limit generated by an admissible finite-energy Type I singularity (u, p, z_*) if it arises from the blow-up construction of [Theorem 1.1](#) along an admissible concentration sequence and satisfies the local L^3 nonvanishing condition [\(1.4\)](#).

Definition 5.8 (Collection of normalized Seregin limits). A collection $\mathcal{S}$ is any nonempty set of normalized Seregin ancient limits generated by admissible finite-energy Type I singularities. When needed, $\mathcal{S}(u, p, z_*)$ denotes a collection associated with a fixed singular point z_* .

Remark 5.9. The local L^3 nonvanishing condition is part of the normalization, not an additional hypothesis. It is inherited from the selected velocity concentration and the no-escape condition through [Theorems 4.1](#) and [4.2](#). Thus every normalized Seregin ancient limit is nonzero in the sense of [\(1.4\)](#).

Proof of Theorem 1.1. Choose an admissible concentration sequence $\lambda_k \downarrow 0$ from [Theorem 2.4](#), and form the rescaled sequence [\(3.2\)](#). The compact ancient limit is [Theorem 3.10](#); the strong and weak convergence are [Theorem 3.6](#). The no-escape argument in [Theorem 4.1](#), followed by strong L^3 -compactness away from $t = 0$, gives nontriviality through [Theorem 4.2](#). The pointwise Type I bound is [Theorem 5.3](#), and the centered equation, smoothness, boundedness, and local L^3 nonvanishing are [Theorems 5.1, 5.2](#) and [5.4](#). $\square$

6. LIOUVILLE CLASSES AND THE REMAINING CLASS

Fix a collection $\mathcal{S}$ of normalized Seregin ancient limits. Complements below are taken relative to $\mathcal{S}$. Structural conditions involving the unrenormalized ancient field refer to

$$U(x, t) = (-t)^{-1/2} V(x/\sqrt{-t}, -\log(-t)).$$

Definition 6.1 (Small-amplitude class). Let $\varepsilon_0 > 0$ be the small-data Liouville constant for bounded ancient solutions of (1.3). We say $V \in \mathcal{C}_{\text{small}}$ if

$$\|V\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})} \leq \varepsilon_0.$$

Definition 6.2 (Stationary L^3 class). We say $V \in \mathcal{C}_{\text{stat}, L^3}$ if

$$V(y, \tau) = W(y) \quad \text{and} \quad W \in L^3(\mathbb{R}^3).$$

Definition 6.3 (Uniformly L^3 -tight class). We say $V \in \mathcal{C}_{L^3\text{-tight}}$ if

$$(6.1) \quad \sup_{\tau \in \mathbb{R}} \|V(\tau)\|_{L^3(\mathbb{R}^3)} < \infty$$

and

$$(6.2) \quad \lim_{R \rightarrow \infty} \sup_{\tau \in \mathbb{R}} \int_{|y| > R} |V(y, \tau)|^3 dy = 0.$$

Definition 6.4 (Liouville-covered structure and decay class). A normalized Seregin limit V belongs to $\mathcal{C}_{\text{known}}$ if its physical ancient representative U satisfies at least one of the following analytic hypotheses:

- (i) U is axisymmetric without swirl;
- (ii) U is axisymmetric and satisfies the pointwise scale-invariant bound

$$|U(x, t)| \leq \frac{C}{r}$$

for some $C < \infty$, where r is the cylindrical radius;

- (iii) U is axisymmetric and

$$rU_\theta \in L_t^\infty L_x^p(\mathbb{R}^3 \times (-\infty, 0))$$

for some $1 \leq p < \infty$;

- (iv) U is axisymmetric, periodic in the axial variable, and

$$rU_\theta \in L^\infty(\mathbb{R}^3 \times (-\infty, 0));$$

- (v) the associated rescaled vorticity satisfies the perturbative Gallay–Wayne weighted-vorticity hypotheses used in [5];
- (vi) U has bounded circulation $\Gamma = rU_\theta$ and Γ satisfies the coercive shell-control hypotheses in the scalar flattening theorem of [5].

Remark 6.5. Only hypotheses for which a cited Liouville theorem gives $U \equiv 0$ are included in $\mathcal{C}_{\text{known}}$. Broader qualitative descriptions such as controlled swirl, weighted velocity decay, Burgers-vortex asymptotics, or bounded nonperiodic circulation are not included unless they imply one of the listed hypotheses.

Lemma 6.6 (Structural quantities under centering). *The centered change of variables preserves axisymmetry and absence of swirl. If (r, z, t) and (ρ, Z, τ) are related by*

$$\rho = \frac{r}{\sqrt{-t}}, \quad Z = \frac{z}{\sqrt{-t}}, \quad \tau = -\log(-t),$$

then the swirl variables satisfy

$$\rho V_\theta(\rho, Z, \tau) = r U_\theta(r, z, t).$$

Proof. The map $x \mapsto y = x/\sqrt{-t}$ is radial in the cylindrical variables and does not mix the e_r, e_θ, e_z frame. Since $V(y, \tau) = \sqrt{-t} U(x, t)$, the θ -components obey $V_\theta = \sqrt{-t} U_\theta$, and the displayed identities follow. $\square$

Definition 6.7 (Remaining class). The remaining class is

$$\mathcal{R}(\mathcal{S}) = \mathcal{S} \setminus (\mathcal{C}_{\text{small}} \cup \mathcal{C}_{\text{stat}, L^3} \cup \mathcal{C}_{L^3\text{-tight}} \cup \mathcal{C}_{\text{known}}).$$

Remark 6.8. The first four classes are not intended to be disjoint. The remaining class is the complement of their union, so overlaps among the Liouville classes cause no ambiguity.

Proposition 6.9 (Exhaustion by defined classes). *Every $V \in \mathcal{S}$ belongs either to one of the four Liouville classes or to $\mathcal{R}(\mathcal{S})$.*

Proof. Let

$$\mathcal{U} = \mathcal{C}_{\text{small}} \cup \mathcal{C}_{\text{stat}, L^3} \cup \mathcal{C}_{L^3\text{-tight}} \cup \mathcal{C}_{\text{known}}.$$

For $V \in \mathcal{S}$, either $V \in \mathcal{U}$, in which case it lies in at least one Liouville class, or $V \in \mathcal{S} \setminus \mathcal{U} = \mathcal{R}(\mathcal{S})$. $\square$

7. LIOUVILLE HYPOTHESES AND CLASS EXCLUSIONS

Lemma 7.1 (Centered Stokes kernel estimates). *Let*

$$L = \Delta - \frac{1}{2}y \cdot \nabla - \frac{1}{2}, \quad S(s) = e^{sL}.$$

There is a universal constant C_0 such that, for all $s > 0$,

$$\|S(s)f\|_{L^\infty} \leq e^{-s/2} \|f\|_{L^\infty}$$

and

$$\|S(s)\mathbb{P}\nabla \cdot F\|_{L^\infty} \leq C_0 e^{-s/2} (1 - e^{-s})^{-1/2} \|F\|_{L^\infty}.$$

Proof. The first estimate follows from the explicit Ornstein–Uhlenbeck kernel:

$$S(s)f(y) = e^{-s/2} \int_{\mathbb{R}^3} G_{1-e^{-s}}(e^{-s/2}y - z)f(z) dz,$$

where G_a is the heat kernel with variance a . The kernel has total mass one. For the projected-divergence estimate, write the corresponding kernel as the Oseen projected-gradient kernel after the same change of variables. More explicitly, the semigroup for $\Delta - \frac{1}{2}y \cdot \nabla$ is

$$(e^{s(\Delta - \frac{1}{2}y \cdot \nabla)}g)(y) = (e^{(1-e^{-s})\Delta}g)(e^{-s/2}y).$$

Applying this to the distribution $g = \mathbb{P}\nabla \cdot F$ and using the projected-gradient Oseen kernel estimate

$$\|e^{\theta\Delta}\mathbb{P}\nabla \cdot F\|_\infty \leq C\theta^{-1/2}\|F\|_\infty, \quad \theta > 0,$$

gives the displayed estimate with $\theta = 1 - e^{-s}$, together with the extra factor $e^{-s/2}$ from the zeroth-order term in L . The projected-gradient Oseen estimate follows from the L^1 -bound

$$\|\nabla O(\cdot, \theta)\|_{L^1} \leq C\theta^{-1/2}$$

for the gradient of the Oseen kernel $O(\cdot, \theta)$. This is the bounded-data Stokes estimate used by Kato [8]; it does not require boundedness of the Leray projector itself on L^∞ . $\square$

Lemma 7.2 (Mild representation for normalized limits). *Every normalized Seregin ancient limit satisfies, for all $\Theta > 0$,*

$$V(\tau) = S(\Theta)V(\tau - \Theta) - \int_0^\Theta S(\Theta - \sigma)\mathbb{P}\nabla \cdot (V \otimes V)(\tau - \Theta + \sigma) d\sigma.$$

Proof. Normalized Seregin limits are smooth and bounded by [Theorems 5.2](#) and [5.4](#). On a finite interval $[\tau - \Theta, \tau]$, the tensor $V \otimes V$ is bounded, so [Theorem 7.1](#) makes the Duhamel integral finite in L^∞ . In the sense of solenoidal distributions, applying the Leray projector to [\(1.3\)](#) removes the pressure and gives

$$\partial_\tau V = LV - \mathbb{P}\nabla \cdot (V \otimes V)$$

on every finite interval. Since V is smooth, the map

$$s \mapsto S(\tau - s)V(s)$$

is differentiable as a tempered distribution for $\tau - \Theta < s < \tau$, and

$$\frac{d}{ds}(S(\tau - s)V(s)) = -S(\tau - s)\mathbb{P}\nabla \cdot (V \otimes V)(s).$$

Integrating this identity from $\tau - \Theta$ to τ gives the stated Duhamel formula. $\square$

Lemma 7.3 (Small bounded ancient Liouville theorem). *There is $\varepsilon_0 > 0$ such that every bounded mild ancient solution of [\(1.3\)](#) with*

$$\|V\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})} \leq \varepsilon_0$$

is identically zero.

Proof. Write the projected centered equation as

$$\partial_\tau V = LV - \mathbb{P}\nabla \cdot (V \otimes V), \quad L = \Delta - \frac{1}{2}y \cdot \nabla - \frac{1}{2}.$$

Let $S(s) = e^{sL}$. By [Theorem 7.1](#), the nonlinear kernel is integrable on $(0, \infty)$; set

$$A = C_0 \int_0^\infty e^{-s/2}(1 - e^{-s})^{-1/2} ds < \infty.$$

For fixed τ and $T > 0$, the mild identity on $[\tau - T, \tau]$ gives

$$V(\tau) = S(T)V(\tau - T) - \int_0^T S(T - \sigma)\mathbb{P}\nabla \cdot (V \otimes V)(\tau - T + \sigma) d\sigma.$$

If $M = \|V\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})}$, then

$$\|V(\tau)\|_\infty \leq e^{-T/2}M + AM^2.$$

Letting $T \rightarrow \infty$ and taking the supremum in τ gives

$$M \leq AM^2.$$

If $0 < M < 1/A$, this is impossible. Choose $\varepsilon_0 < 1/A$; then $M \leq \varepsilon_0$ implies $M = 0$. $\square$

Lemma 7.4 (Stationary centered profiles). *If $V(y, \tau) = W(y)$ is a smooth solution of [\(1.3\)](#), then after subtracting a function of τ from the pressure, W solves*

$$(W \cdot \nabla)W + \nabla P = \Delta W - \frac{1}{2}(W + y \cdot \nabla W), \quad \operatorname{div} W = 0.$$

Proof. Since V is independent of τ , $\partial_\tau V = 0$. Set

$$F(y) = \Delta W - \frac{1}{2}(W + y \cdot \nabla W) - (W \cdot \nabla)W.$$

The centered equation gives $\nabla_y \Pi(\cdot, \tau) = F$ for a.e. τ . Fix one such time τ_0 and put $P(y) = \Pi(y, \tau_0)$. Then $\nabla_y(\Pi(\cdot, \tau) - P) = 0$ for a.e. τ , so $\Pi(y, \tau) - P(y) = b(\tau)$ is spatially constant. Subtracting this gauge from Π leaves the equation unchanged and yields the displayed stationary self-similar equation for W . $\square$

Theorem 7.5 (Small and stationary class exclusions). *Let V be a normalized Seregin ancient limit. Then*

$$V \notin \mathcal{C}_{\text{small}}, \quad V \notin \mathcal{C}_{\text{stat}, L^3}.$$

Proof. If $V \in \mathcal{C}_{\text{small}}$, then [Theorems 7.2](#) and [7.3](#) give $V \equiv 0$, contradicting the local L^3 nonvanishing condition [\(1.4\)](#).

If $V \in \mathcal{C}_{\text{stat}, L^3}$, then $V(y, \tau) = W(y)$ with $W \in L^3(\mathbb{R}^3)$. By [Theorem 7.4](#), W satisfies the stationary self-similar profile equation after a pressure gauge change. The theorem of Nečas–Růžička–Šverák [\[15\]](#) gives $W \equiv 0$. Again [\(1.4\)](#) gives the contradiction. $\square$

Theorem 7.6 (Uniformly tight L^3 closure). *No nonzero normalized Seregin ancient limit belongs to $\mathcal{C}_{L^3\text{-tight}}$.*

Proof. This is the tight-class Liouville theorem proved in Paper II [\[6, Corollary Tight-class input for the residual criterion\]](#). Paper II applies the physical pullback and the Albritton–Barker endpoint ancient Liouville theorem to uniformly L^3 -tight centered ancient profiles carrying the Paper I compact velocity normalization. $\square$

Theorem 7.7 (Liouville-covered structure and decay exclusions). *No nonzero normalized Seregin ancient limit belongs to $\mathcal{C}_{\text{known}}$.*

Proof. By [Theorem 6.4](#), membership in $\mathcal{C}_{\text{known}}$ means that the physical ancient representative U satisfies one of the listed analytic hypotheses. The axisymmetric no-swirl and pointwise $1/r$ cases are excluded by Koch–Nadirashvili–Seregin–Šverák [\[9\]](#). The finite L^p circulation case is excluded by Lei–Zhang–Zhao [\[14\]](#), and the axial-periodic bounded-circulation case is excluded by Lei–Ren–Zhang [\[13\]](#). The perturbative Gallay–Wayne weighted-vorticity hypotheses and the bounded-circulation shell-control hypotheses are excluded by the corresponding Liouville theorems in Paper III [\[5\]](#). Each alternative gives $V \equiv 0$, contradicting the local L^3 nonvanishing condition [\(1.4\)](#). Hence no nonzero normalized Seregin ancient limit lies in $\mathcal{C}_{\text{known}}$. $\square$

Proposition 7.8 (Exclusion of the Liouville classes). *No normalized Seregin ancient limit can belong to*

$$\mathcal{C}_{\text{small}} \cup \mathcal{C}_{\text{stat}, L^3} \cup \mathcal{C}_{L^3\text{-tight}} \cup \mathcal{C}_{\text{known}}.$$

Proof. The small and stationary L^3 classes are excluded by [Theorem 7.5](#). The uniformly tight class is excluded by [Theorem 7.6](#). The structure/decay class is excluded by [Theorem 7.7](#). $\square$

8. THE REMAINING-CASE CRITERION

Theorem 8.1 (Residual closure theorem). *For every collection $\mathcal{S}$ of normalized Seregin ancient limits generated by admissible finite-energy Type I singularities,*

$$\mathcal{R}(\mathcal{S}) = \emptyset.$$

Proof. This is the Paper I residual input proved in the residual paper [7, Corollary *Paper I residual input*]. The residual paper uses the same Paper I class exhaustion. Its notation $\mathcal{R}(\mathcal{S}; I, J)$ records the ordered removal of the small, stationary L^3 , uniformly L^3 -tight, and Liouville-covered structure/decay classes, together with the refined lower residual classes. No additional analytic hypothesis is introduced by the indices I, J . Its four local exclusions give $\mathcal{R}(\mathcal{S}; I, J) = \emptyset$, which is precisely the emptiness of the Paper I remaining class $\mathcal{R}(\mathcal{S})$. $\square$

Corollary 8.2 (Rigidity in the remaining case). *For every collection $\mathcal{S}$ of normalized Seregin ancient limits generated by admissible finite-energy Type I singularities, every element of $\mathcal{R}(\mathcal{S})$ is identically zero.*

Proof. This is immediate from Theorem 8.1. $\square$

Proposition 8.3 (Equivalence on normalized Seregin limits). *For collections of normalized Seregin limits, Theorem 8.1 and Theorem 8.2 are equivalent.*

Proof. If Theorem 8.1 holds, the rigidity statement is vacuous. Conversely, assume Theorem 8.2. If some normalized collection had $V \in \mathcal{R}(\mathcal{S})$, the rigidity hypothesis would give $V \equiv 0$, directly contradicting the local L^3 nonvanishing condition (1.4). Hence the remaining class is empty. $\square$

Proof of Theorem 1.2. Suppose that (u, p, z_*) is an admissible finite-energy Type I singularity. By Theorem 1.1, there exists a normalized nonzero Seregin ancient limit V . Let $\mathcal{S}$ be any collection of normalized limits containing V .

By Theorem 6.9, either V belongs to one of the Liouville classes or $V \in \mathcal{R}(\mathcal{S})$. The first alternative is impossible by Theorem 7.8. The second alternative is impossible by Theorem 8.1. Hence the assumed admissible Type I singularity cannot exist. $\square$

The preceding theorem is the unconditional Type I conclusion for the stated admissible finite-energy class. The compactness construction, pressure normalization, and class exhaustion are contained in this paper, with the local velocity concentration input stated explicitly in Theorem 2.7. The structure/decay class is restricted to the assumptions in Theorem 6.4; any other structured behavior lies in the remaining class and is covered by Theorem 8.1. No whole-space critical-norm estimate on the original solution is used in this closure step.

APPENDIX A. ELEMENTARY SYMMETRIES OF THE CENTERED EQUATION

Lemma A.1 (Logarithmic-time translation). *If V solves (1.3), then*

$$V_{\tau_0}(y, \tau) = V(y, \tau + \tau_0)$$

also solves (1.3). If

$$U^\lambda(x, t) = \lambda U(\lambda x, \lambda^2 t),$$

then the renormalized field associated with U^λ is

$$V^\lambda(y, \tau) = V(y, \tau - 2 \log \lambda).$$

Proof. The first statement follows because (1.3) is autonomous in τ . For the second, put $s = -t = e^{-\tau}$. Then

$$\sqrt{s} U^\lambda(y\sqrt{s}, -s) = \lambda\sqrt{s} U(\lambda y\sqrt{s}, -\lambda^2 s) = V(y, -\log(\lambda^2 s)) = V(y, \tau - 2 \log \lambda).$$

$\square$

Proposition A.2 (Failure of generic dilation and translation symmetries). *The centered equation (1.3) is autonomous in τ , but it is not invariant under*

$$V(y, \tau) \mapsto \alpha V(\alpha y, \alpha^2 \tau)$$

unless $\alpha = 1$. It is also not invariant under constant translations

$$V(y, \tau) \mapsto V(y - a, \tau)$$

unless $a = 0$.

Proof. For scaling, set $Y = \alpha y$, $T = \alpha^2 \tau$, and $\tilde{V}(y, \tau) = \alpha V(Y, T)$, $\tilde{\Pi}(y, \tau) = \alpha^2 \Pi(Y, T)$. The non-drift part scales with factor α^3 , while

$$-\frac{1}{2}(\tilde{V} + y \cdot \nabla \tilde{V}) = -\frac{\alpha}{2}(V + Y \cdot \nabla_Y V)(Y, T).$$

Using the equation for V , the equation defect for $\tilde{V}$ is

$$\frac{\alpha - \alpha^3}{2}(V + Y \cdot \nabla_Y V)(Y, T).$$

If $\alpha \neq 1$, this forces $V + Y \cdot \nabla V = 0$. Along each ray,

$$\frac{d}{dr}(rV(r\omega, T)) = 0.$$

Boundedness and smoothness at the origin force the constant to be zero, hence $V \equiv 0$. Thus a nonzero centered profile has no such scaling symmetry.

For translations, all translation-invariant terms transform by composition, but the drift satisfies

$$y \cdot \nabla V(y - a, \tau) = (y - a) \cdot \nabla V(y - a, \tau) + a \cdot \nabla V(y - a, \tau).$$

The extra term is absent from the original equation. Thus spatial translation is not a symmetry of the fixed centered equation unless $a = 0$, apart from exceptional fixed solutions satisfying $a \cdot \nabla V \equiv 0$. $\square$

REFERENCES

- [1] D. Albritton and T. Barker, *On local Type I singularities of the Navier–Stokes equations and Liouville theorems*, J. Math. Fluid Mech. **21** (2019), article no. 43, doi:<https://doi.org/10.1007/s00021-019-0448-z>.
- [2] J.-P. Aubin, *Un theoreme de compacite*, C. R. Acad. Sci. Paris **256** (1963), 5042–5044.
- [3] L. Caffarelli, R. Kohn, and L. Nirenberg, *Partial regularity of suitable weak solutions of the Navier–Stokes equations*, Comm. Pure Appl. Math. **35** (1982), no. 6, 771–831, doi:<https://doi.org/10.1002/cpa.3160350604>.
- [4] L. Escauriaza, G. Seregin, and V. Šverak, *$L_{3,\infty}$ -solutions of Navier–Stokes equations and backward uniqueness*, Russian Math. Surveys **58** (2003), no. 2, 211–250, doi:<https://doi.org/10.1070/RM2003v058n02ABEH000609>.
- [5] G. Duran Ballester, *Structured ancient solution exclusions for Type I Navier–Stokes blow-up limits*, preprint, 2026.
- [6] G. Duran Ballester, *Uniformly tight ancient Navier–Stokes dynamics and endpoint L^3 rigidity*, preprint, 2026.
- [7] G. Duran Ballester, *Local exclusion of residual Type I Navier–Stokes Seregin limits*, preprint, 2026.
- [8] T. Kato, *Strong L^p -solutions of the Navier–Stokes equation in $\mathbb{R}^m$, with applications to weak solutions*, Math. Z. **187** (1984), no. 4, 471–480, doi:<https://doi.org/10.1007/BF01174182>.
- [9] G. Koch, N. Nadirashvili, G. Seregin, and V. Šverak, *Liouville theorems for the Navier–Stokes equations and applications*, Acta Math. **203** (2009), no. 1, 83–105.
- [10] F.-H. Lin, *A new proof of the Caffarelli–Kohn–Nirenberg theorem*, Comm. Pure Appl. Math. **51** (1998), no. 3, 241–257, doi:[https://doi.org/10.1002/\(SICI\)1097-0312\(199803\)51:3<241::AID-CPA2>3.0.CO;2-A](https://doi.org/10.1002/(SICI)1097-0312(199803)51:3<241::AID-CPA2>3.0.CO;2-A).
- [11] J. Leray, *Sur le mouvement d’un liquide visqueux emplissant l’espace*, Acta Math. **63** (1934), 193–248.

- [12] J.-L. Lions and E. Magenes, *Non-homogeneous boundary value problems and applications. Vol. I*, Springer-Verlag, 1972.
 - [13] Z. Lei, X. Ren, and Q. S. Zhang, *A Liouville theorem for the axially symmetric Navier–Stokes equations*, J. Funct. Anal. **283** (2022), no. 3, 109522.
 - [14] Z. Lei, Q. Zhang, and N. Zhao, *Improved Liouville theorems for axially symmetric Navier–Stokes equations*, Sci. Sin. Math. **47** (2017), no. 10, 1183–1198.
 - [15] J. Necas, M. Ruzicka, and V. Šverák, *On Leray’s self-similar solutions of the Navier–Stokes equations*, Acta Math. **176** (1996), no. 2, 283–294.
 - [16] G. Duran Ballester, *Local CKN concentration and the Type I/Type II dichotomy for the Navier–Stokes equations*, preprint, 2026.
 - [17] G. Seregin, *A certain necessary condition of potential blow up for Navier–Stokes equations*, Comm. Math. Phys. **312** (2012), no. 3, 833–845, doi:<https://doi.org/10.1007/s00220-011-1391-x>.
 - [18] J. Serrin, *On the interior regularity of weak solutions of the Navier–Stokes equations*, Arch. Rational Mech. Anal. **9** (1962), 187–195, doi:<https://doi.org/10.1007/BF00253344>.
 - [19] J. Simon, *Compact sets in the space $L^p(0, T; B)$* , Ann. Mat. Pura Appl. (4) **146** (1987), 65–96, doi:<https://doi.org/10.1007/BF01762360>.
 - [20] H. Sohr, *The Navier–Stokes equations. An elementary functional analytic approach*, Birkhauser, 2001.
 - [21] E. M. Stein, *Singular integrals and differentiability properties of functions*, Princeton University Press, 1970.
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